FIELD REPORT / 23 JULY 2026

Intelligent Terrarium

What we built on **jas-nzxt**, what the screen actually was, how much mind fits inside 1–2 GB of RAM, and how this could become a personal computer that grows a strange history of its own.

EXPERIMENT CLOSED CLEANLY

ISOLATED BRANCH agent/intelligent-terrarium
STATE REPOSITORY /home/me/intelligent-terrarium-state
STATUS kiosk stopped · authority absent · miner active

THE ANSWER IN ONE SENTENCE

A real bounded mind was prototyped; the beautiful fullscreen creature was not yet connected to it.

The work separated world truth from spectacle. That was technically correct, but it also meant the final screen was procedural theater: lively, persistent-looking, and visually suggestive—yet not evidence that the terrarium was thinking.

26/26	912	0
LOCAL TESTS PASSED	FINAL JOURNAL SEQUENCE	LIVE TERRARIUM PROCESSES NOW

REAL AND TESTED

Deterministic simulation, bounded episodic memory, semantic sound events, hash-chained journal/replay, crash-tail quarantine, Git sleep commits, identity seam, loopback visitor, and a constrained reflection organ.

NOT YET REAL

An always-on authority, live telemetry driving the display, physical Xbox QA, a public authenticated gateway, multi-user spatial presence, and continuous autonomous growth.

MEASURED ON JAS-NZXT

1 GB fallback stayed tiny. A 2 GB run loaded Qwen3 0.6B Q8_0 CPU-only, peaked near 807 MiB RSS for the inference child, used no swap, and coexisted with the GPU miner after its automatic recovery.

THE SCREEN'S DECEPTION

The 82,944 voxels, amoeboid organs, traces, "culture age," and numeric soup were generated by browser time functions and local storage. They did not read a live mind or operating-system telemetry.

WHY "84K" WAS BOUNDED

That number was approximately the fixed **48 × 36 × 48 voxel lattice**—a rendering budget, not the terrarium's memory or intelligence. Bounded memory is a safety property: old experience is compressed into summaries and weights so the organism can accumulate history without eventually consuming the machine.

SYSTEM ANATOMY

One authority, one touchable membrane, many renderers.

The core architectural choice is strong: the browser is a visitor, never the source of truth. The authority receives verified intentions, advances a deterministic world, emits snapshots and sonic events, and periodically sleeps into a separate Git autobiography.

Visitors → **Mediorgan** → **terrariumd**

Visitors
Web / AC piece / keyboard / gamepad / future Xbox

Mediorgan
loopback gate · identity · rate bounds · causal prods

terrariumd
single writer · deterministic policy · bounded world

↓

Semantic sound ← **Snapshots** → **Git sleep**

Semantic sound
source, pitch, voice, radius, cause—not an audio stream

Snapshots
positions · drives · behaviors · measurements

Git sleep
journal · checkpoint · autobiography · replay hash

THE BOUNDED ORGANS

SENSORY · SPATIAL · DRIVE · MEMORY · ACTION · VOICE · SLEEP

DURABLE SUBSTRATE

- Canonical NDJSON facts with sequence and hash chain
- Atomic checkpoints that are verified by replay
- Bounded human-readable autobiographies
- Local Git commits on sleep; no automatic push

SAFETY BOUNDARY

- Loopback-only authority in the prototype
- One writer for state and journal ordering
- Tokens, emails, IPs, model caches, and raw memories excluded
- Hard cgroup RAM and zero-swap profiles

The working implementation lives under **experiments/intelligent-terrarium/** on an isolated branch. The autobiographical state lives separately on the tower so autonomous commits cannot trigger Aesthetic Computer deployment hooks.

RESIDENT INTELLIGENCE BUDGETS

One gigabyte is enough for a creature. Two is enough for a creature that occasionally reflects in language.

1 GB profile

PRODUCT FLOOR

measured core peak ≈ 32 MB cgroup

Deterministic drives, spatial graph, bounded episodes, salience/novelty, learned weights, behavior policy, semantic sound, journaling, and replay. No resident language model. Most of the budget remains safety headroom.

2 GB profile

MEASURED REFLECTION

Qwen child peak ≈ 807 MiB RSS

Everything above plus one CPU-only Qwen3 0.6B Q8_0 reflection request at a time: bounded prompt, 2,048-token context cap, short output, strict parsing, and deterministic accept/reject/fallback.

639 MB	0	4	1
MODEL FILE	SWAP USED	CPU THREADS	INFERENCE AT ONCE

WHAT CREATES INTELLIGENCE HERE

Continuity, perception, drives, memory selection, prediction, action, social consequence, and replayable learning. Parameter count helps with language, but it is not the same thing as having a life.

WHERE TINY MODELS BELONG

As slow organs: naming, dreaming, summarizing, proposing, or translating internal state into language. They should never be the authority and never be allowed to rewrite their own history unchecked.

MY PRACTICAL JUDGMENT

A 1 GB machine can host a compelling persistent artificial organism if its intelligence is **architectural**, not chatbot-shaped. At 2 GB, a sub-billion-parameter model can add linguistic texture. Trying to cram a much larger general assistant into the same envelope would sacrifice responsiveness, safety headroom, or both.

GROWTH WITHOUT RAM CREEP

The terrarium should grow by changing its history and phenotype—not by keeping everything resident.

A bounded organism can become stranger for years if yesterday changes how today is perceived. Long duration is a compression problem, not a hoarding problem.

Wake replay last verified state › **Perceive** world, visitors, machine weather › **Act** behavior, sound, growth › **Digest** retain novelty; compress repetition › **Sleep** verify, narrate, Git commit

FAST LIFE	10 Hz simulation, movement, impulses, oscillators, energy flow, local sound. It is ephemeral and cheap.
WORKING MEMORY	A few hundred recent and salient episodes, bounded queues, current relationships, and active drives.
LONG MEMORY	Hierarchical summaries, learned dispositions, named places, visitor relationships, recurring motifs, and hashes back to source events.
PHENOTYPE	A compact “genome” of weights, thresholds, palettes, rhythmic tendencies, growth rules, and scars. Small mutations create visible divergence.
AUTOBIOGRAPHY	Git commits as sleep epochs: diffable, branchable, inspectable, reversible, and shareable without model weights.

GOOD GNARLINESS

New habits, grudges, affinities, routes, timbres, morphology, seasonal cycles, fossils, and myths that can be traced to experience.

BAD GNARLINESS

Unbounded logs, self-rewriting authority, prompt accumulation, random visual noise, uncontrolled network access, or a dashboard pretending to measure cognition.

Key design move: batch or summarize hot simulation before journaling. Writing every 100 ms forever would create roughly 864,000 records per day and turn Git into a landfill. A real daemon should record meaningful transitions or minute-scale digests, then make hour/day-scale sleep commits.

A CREDIBLE RESURRECTION

Make the screen truthful first. Then make the terrarium visitable.

1 · Continuous local authority

Run a supervised loopback-only daemon under the 1 GB cgroup. Journal meaningful transitions at a slow bounded cadence. Expose read-only live telemetry to the display.

2 · A display that is the organism

Every voxel field, organ shape, number, and sound must derive from actual drives, events, episodes, mutations, and system measurements. No decorative metrics presented as cognition.

3 · Ecological memory

Add energy, metabolism, territories, reproduction, inheritance, mutation, and a compact phenotype. Let visitors leave consequences that survive sleep and replay.

4 · AC handles and spatial-sonic visitors

Complete the verified token → Auth0 subject → current AC handle path. Ship a hidden web client with keyboard/gamepad navigation, semantic WebAudio, and an observer/interactor access distinction.

5 · Xbox and remote gateway

Only after threat review: TLS gateway, bounded sessions, restore drills, handle policy, Edge/Xbox testing, and public spectatorship without exposing the authority directly.

SPATIAL

Walk through regions that have history. Near/far and left/right sound comes from authoritative coordinates. The camera is a body, not a free dashboard.

SOCIAL

Handled visitors become remembered relationships. Consent and rate limits decide who may observe, prod, teach, or alter the ecology.

GENERATIVE

The terrarium can emit paintings, scores, stories, maps, seeds, or playable scenes as metabolic by-products with causal links back to lived events.

WHAT I THINK IS POSSIBLE

Not a miniature omniscient assistant. Something more interesting: a small, continuous, situated intelligence whose limits become its character—an inspectable machine ecology that remembers visitors, mutates its audiovisual body, dreams in tiny language bursts, and can be forked from its own Git past.

Closed state, 23 July 2026: `terrarium-score-0723.service` inactive; no process matching the terrarium runtime; no listeners on 8787 or 18787. The unrelated system-level `btx-miner.service` remains active. No branch, state repository, or model was pushed or publicly deployed.